

Day_12_VQE

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WEEK-03: DAY 12: VQE ALGORITHM : RAJIV KUMAR YADAV

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[2]: import numpy as np

from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from scipy.optimize import minimize

# =====
# 1. Hamiltonian  $H = Z$ 
# =====

Z = np.array([
    [1, 0],
    [0, -1]
], dtype=complex)

# =====
# 2. Parameterized quantum circuit
# =====

def create_ansatz(theta):

    # scipy gives theta as numpy array
    theta = float(theta[0])

    qc = QuantumCircuit(1)

    qc.ry(theta, 0)

    return qc

# =====
# 3. Calculate expectation value  $\langle Z \rangle$ 
# =====
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def expectation_value(theta):

    qc = create_ansatz(theta)

    simulator = AerSimulator(method="statevector")

    qc.save_statevector()

    result = simulator.run(qc).result()

    state = result.get_statevector()

    state = np.asarray(state)

    #  $\langle \psi | Z | \psi \rangle$ 
    energy = np.real(
        np.vdot(state, Z @ state)
    )

    return energy

# =====
# 4. Initial parameter
# =====

initial_theta = [0.5]

# =====
# 5. Classical optimizer
# =====

result = minimize(
    expectation_value,
    initial_theta,
    method="COBYLA"
)

# =====
# 6. Results
# =====

print("\n=====")
print("          VQE RESULT")

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print("=====")

print("Optimal theta :", result.x[0])
print("Minimum energy:", result.fun)

print("Exact energy  :", -1.0)

print("=====")

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                VQE RESULT
=====
Optimal theta : 3.1416000000000004
Minimum energy: -0.999999999973015
Exact energy  : -1.0
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